Exploratory Data Analysis (EDA): A study of unstructured document data relating to an Irish Retail Bank

Programme: PGDip Applied Data Analytics
Course: SWRDC9011 Machine Learning and AI with Industrial Applications
Lecturer: Dr A. Kaushik
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Agenda

1. **Approach**
2. **Report**
3. **Answers to Key Questions**

Link to screencast:

ML CA2 Screencast.mp4

1. Approach

Step 1: Identify Theme

Step 2: Locate Suitable Source Data

Step 3: Import Data

Step4: Enrich Metadata

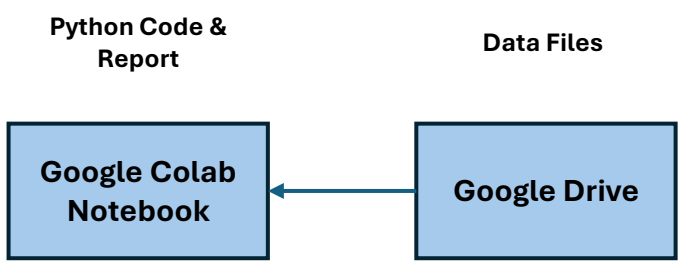
Step 5: Descriptive Statistics

Step 6: Visualisations

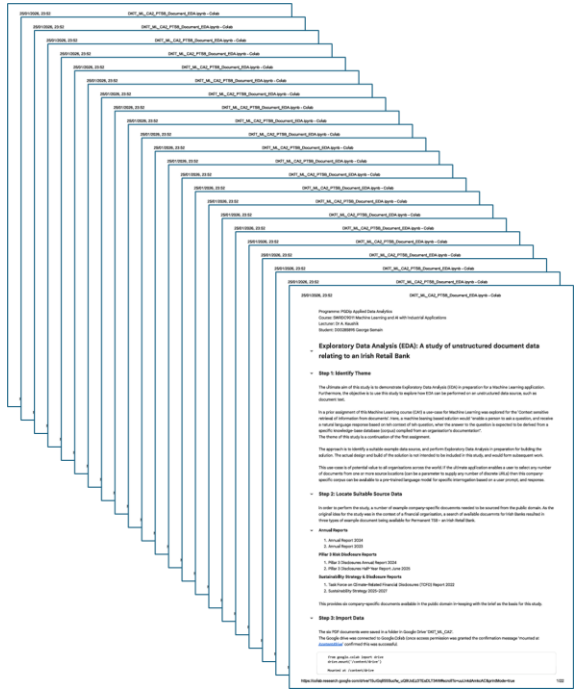
Step 7: Words & Vocabulary Analysis

Step 8: Summary of EDA

References



2. Report



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Step 1: Identify Theme

The ultimate aim of this study is to demonstrate Exploratory Data Analysis (EDA) in preparation for a Machine Learning application. Furthermore, the objective is to use this study to explore how EDA can be performed on an unstructured data source, such as document text.

In a prior assignment of this Machine Learning course (CA1) a use-case for Machine Learning was explored for the 'Context sensitive retrieval of information from documents'. Here, a machine leaning based solution would "enable a person to ask a question, and receive a natural language response based on teh context of teh question, wher the answer to the question is expected to be derived from a specific knowledge-base database (corpus) compiled from an organisation's documentation".
The theme of this study is a continuation of the first assignment.

The approach is to identify a suitable example data source, and perform Exploratory Data Analysis in preparation for building the solution. The actual design and build of the solution is not intended to be included in this study, and would form subsequent work.

This use-case is of potential value to all organisations across the world. If the ultimate application enables a user to select any number of documents from one or more source locations (can be a parameter to supply any number of discrete URLs) then this company-specific corpus can be available to a pre-trained language model for specific interrogation based on a user prompt, and response.

Step 2: Locate Suitable Source Data

In order to perform the study, a number of example company-specific docuemnts needed to be sourced from the public domain. As the original idea for the study was in the context of a financial organisation, a search of available docuemnts for Irish Banks resulted in three types of example document being available for Permanent TSB - an Irish Retail Bank.

Annual Reports

1. Annual Report 2024
2. Annual Report 2023

Pillar 3 Risk Disclosure Reports

1. Pillar 3 Disclosures Annual Report 2024
2. Pillar 3 Disclosures Half-Year Report June 2025

Sustainability Strategy & Disclosure Reports

1. Task Force on Climate-Related Financial Disclosures (TCFD) Report 2022
2. Sustainability Strategy 2025-2027

This provides six company-specific documents available in the public domain in-keeping with the brief as the basis for this study.

Step 3: Import Data

The six PDF documents were saved in a folder in Google Drive 'DKIT_ML_CA2'.
The Google drive was connected to Google.Colab (once access permission was granted the confirmation message 'mounted at /content/drive' confirmed this was successful.

```
from google.colab import drive
drive.mount('/content/drive')

Mounted at /content/drive
```

Answers to Key Questions

1. **What problem does your project address, and why is it important?** Explain the real-world issue, who is affected, and the expected impact of your solution.
2. **How does your prototype or implementation work (code, non-code, or mixed)?** Describe the system architecture, key modules, data flow, models used, and any important implementation details.
3. **What is the novel or original aspect of your idea, and how does it improve existing solutions?** Highlight originality and add bold, clear statements about novelty and advantages over current approaches.
4. **What is the commercialization potential of your idea?** Describe the target market, revenue model, go-to-market approach, and justify a commercialization score or ranking.
5. **What challenges did you face during development, and how did you overcome them?** Discuss technical, ethical, or challenges and the steps you took to address them.

1. What problem does your project address, and why is it important?

A machine leaning based solution would

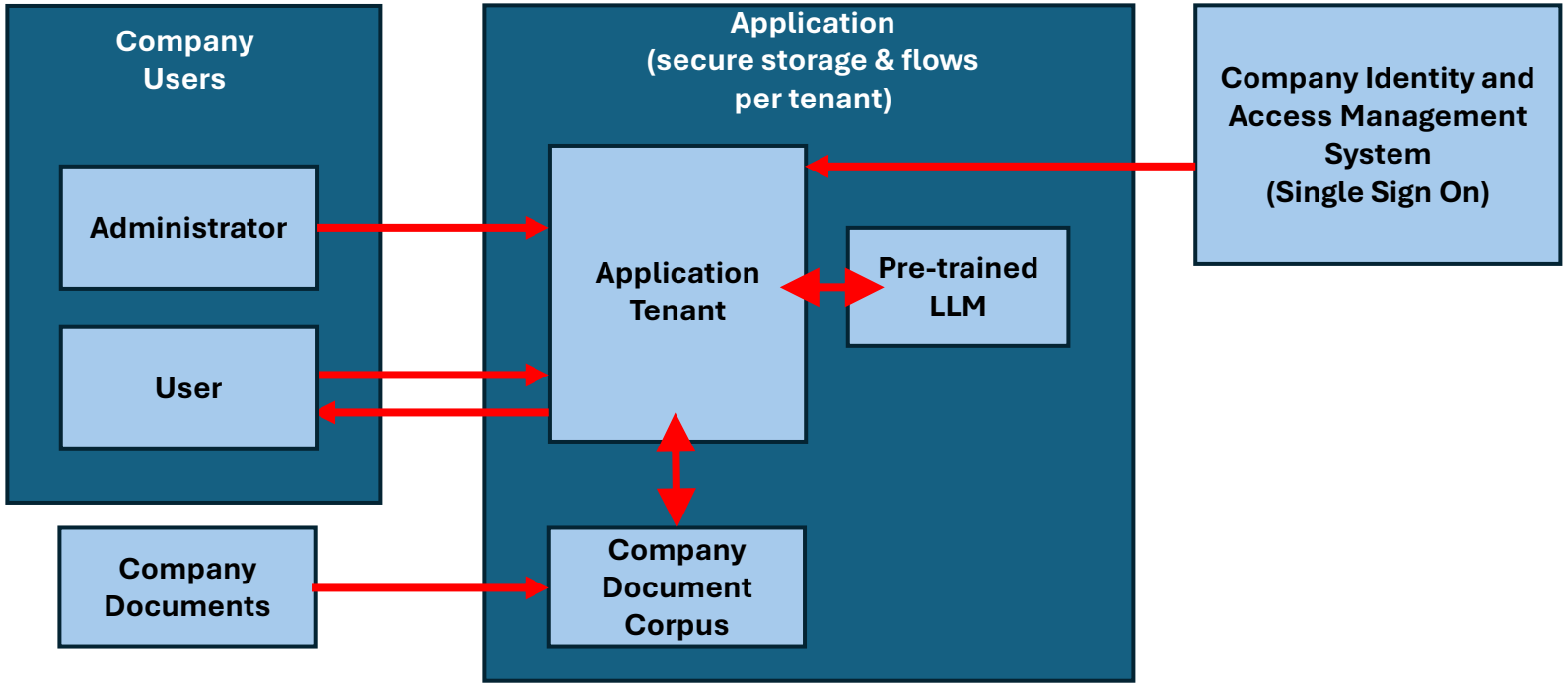
"enable a person to ask a question, and receive a natural language response based on the context of the question, where the answer to the question is expected to be derived from a specific knowledge-base database (corpus) compiled from an organisation's documentation".

This is important as it enables a factual and accurate response to a specific question without the user having to search through potentially lots of documents to find the information they need, or spend the time of subject matter experts for advice. As the corpus is refreshed or augmented, the model will take that into account, always providing the latest details. References to the document(s) and page number(s) can be provided. Additionally, any potentially conflicting information across documents can also be highlighted.

2. How does your prototype or implementation work (code, non-code, or mixed)?

This study covers EDA only and is not a working prototype of the application.

The full implementation would require the following architecture:



3. What is the novel or original aspect of your idea, and how does it improve existing solutions?

The solution enables any company to maintain a relevant corpus themselves, and provide permission to users to a secure tenant through single sign-on.

The aim is that it provides a flexible approach to document management, and securitisation of company information and access control.

4. What is the commercialization potential of your idea?

This use-case is of potential value to **all organisations across the world**.

If the ultimate application enables a user to select any number of documents from one or more source locations (can be a parameter to supply any number of discrete URLs) then this company-specific corpus can be available to a pre-trained language model for specific interrogation based on a user prompt, and response.

5. What challenges did you face during development, and how did you overcome them?

Prior to the study, I was unfamiliar with how to perform Exploratory Data Analysis on unstructured data. Considerable reading from a number of references noted in the report has enabled a step-change in understanding the approach and techniques, albeit still at a foundational level.

Through books and online resources, guidance was required at every stage of the technical implementation in Python, as performing natural language analysis of documents was completely new.